

Adaptive Multi-Timescale CTR Learning with Adaptive Temporal Persistence

Methodology and Experimental Plan

Abstract

This report specifies *Adaptive Multi-Timescale Gating with Temporal Persistence* (AMG-TP), a sequential method for click-through-rate (CTR) prediction under temporal distribution shift. The central objective is to learn, for each incoming example, which historical timescales should be trusted and how rapidly that temporal preference should itself be permitted to change. AMG-TP maintains predictors trained on multiple historical horizons, combines them using a context-dependent gate, and learns a separate persistence parameter that controls the inertia of the deployed horizon mixture. The experimental plan evaluates whether a single causally deployed adaptive model can recover the rapid response of low-persistence gating under abrupt or local drift, retain the stability of high-persistence gating under recurring drift, avoid material degradation under stationary conditions, and translate improved CTR prediction into greater downstream value under an unchanged bidding policy.

Keywords: click-through-rate prediction; temporal distribution shift; online learning; mixture of experts; adaptive forgetting; calibration; autobidding.

1 Empirical Motivation

Current benchmark results indicate that fixed temporal strategies leave measurable performance headroom. On the locked test, expanding-history training is the strongest simple fixed baseline with mean log loss 0.4271, while the best fixed horizon varies across test days among expanding, rolling-3, and rolling-7 configurations. A context-dependent short/long gate (M2) improves the mean log loss to 0.4251. M5b with high smoothness reaches 0.4186; however, its strong temporal persistence is beneficial under recurring drift while being less suitable for abrupt or local drift. These observations motivate adapting both the memory scale and the persistence of the gate.

Table 1: Current locked-test benchmark motivating adaptive memory and adaptive persistence.

Method	Log loss	Interpretation
Expanding	0.4271	Best simple baseline.
Han ARW	0.4291	Adaptive global horizon; strong reference baseline.
M2 context gate	0.4251	Per-example short/long mixing provides an improvement.
M5b high-smooth	0.4186	Best current locked-test mean; particularly strong under recurring drift.
Ensemble3	0.4210	Indicates that regime-specific specialists can be inferred online.

2 Proposed Method: AMG-TP

The provisional method name is **Adaptive Multi-Timescale Gating with Temporal Persistence (AMG-TP)**. The method extends M5b by making temporal persistence adaptive rather than fixing the smoothness strength in advance.

2.1 Temporal Experts

Maintain K CTR predictors trained on distinct historical horizons. The initial expert bank is

$$\mathcal{H} = \{1\text{-day}, 3\text{-day}, 7\text{-day}, 14\text{-day}, \text{expanding}\}. \quad (1)$$

For an incoming example x at time t , expert k outputs $p_t^{(k)}(x)$. Each expert is trained using only information available strictly before the prediction time t .

2.2 Raw Context-Dependent Gate

A reusable gating network proposes a context-dependent distribution over temporal horizons:

$$\mathbf{q}_t(x) = \text{softmax}\left(g_\phi\left(x, p_t^{(1:K)}(x), \mathbf{s}_{t-1}\right)\right). \quad (2)$$

The state $\mathbf{s}_{t-1}$ contains only previously observed information, including recent expert losses, prediction disagreement, feature-shift statistics, recent CTR, and recent gate movement.

2.3 Persistent Temporal State

Maintain a low-dimensional state $\mathbf{m}_{t-1}$ summarizing which horizons have recently been favored by the environment. A simple realization is an exponentially smoothed average of previously deployed gate weights:

$$\mathbf{m}_t = (1 - \rho)\mathbf{m}_{t-1} + \rho \mathbb{E}_x[\boldsymbol{\pi}_t(x)]. \quad (3)$$

2.4 Adaptive Persistence

Instead of fixing a global smoothness coefficient, predict the amount of temporal inertia imposed at time t :

$$\beta_t = \sigma(r_\psi(\mathbf{s}_{t-1})), \quad \beta_t \in [0, 1]. \quad (4)$$

A low value of β_t permits rapid movement after abrupt change, whereas a high value preserves stable memory allocation when the environment is persistent or recurring.

2.5 Deployed Memory Weights and CTR Prediction

The deployed temporal mixture is

$$\boldsymbol{\pi}_t(x) = (1 - \beta_t)\mathbf{q}_t(x) + \beta_t\mathbf{m}_{t-1}, \quad (5)$$

and the final CTR estimate is

$$\hat{p}_t(x) = \sum_{k=1}^K \pi_{t,k}(x) p_t^{(k)}(x). \quad (6)$$

This factorization separates two adaptation decisions: (i) which historical timescale an individual example should trust, represented by $\mathbf{q}_t(x)$ and the deployed $\pi_t(x)$; and (ii) how rapidly the temporal preference itself should change, represented by β_t .

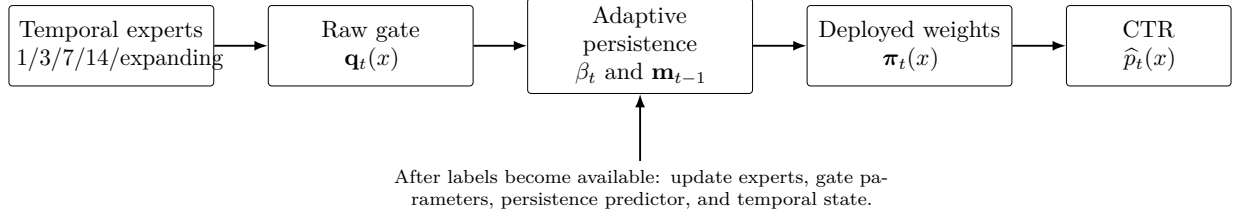


Figure 1: Sequential deployment of AMG-TP. The method is intentionally stateful; arbitrary multi-step-ahead gate prediction is not required for online CTR or autobidding deployment.

2.6 Sequential Deployment Semantics

At a genuinely future time t , the outcome y_t is unknown. AMG-TP therefore computes π_t from the stored state and current observable context, predicts CTR, and updates only after the outcome becomes available:

$$\text{state}_{t-1} \rightarrow \pi_t(x_t) \rightarrow \hat{p}_t(x_t) \rightarrow \text{observe } y_t \rightarrow \text{state}_t. \quad (7)$$

The exact deployed π_{t+h} cannot be obtained by jumping directly to a future step without rolling through intermediate states because those states depend on future observations. This property is consistent with the intended online application, in which CTR prediction proceeds sequentially at impression arrival or temporal-block arrival.

3 Training and Online Update Procedure

The proposed implementation follows the sequence below.

1. Initialize temporal experts, gate parameters ϕ , persistence-model parameters ψ , and state $\mathbf{m}_0$ using the development history.
2. At period t , update each expert using only its permitted historical window ending before t .
3. For each current example, compute all expert predictions and the raw gate $\mathbf{q}_t(x)$.
4. Compute β_t using past-state features only, construct $\pi_t(x)$, and output the mixture CTR prediction.
5. After labels for period t mature, compute a proper prediction loss, with log loss as the primary objective, and update ϕ and ψ by gradient descent or an equivalent online/meta-learning rule.
6. Update $\mathbf{m}_t$ from deployed gate weights and advance to $t + 1$.

Recommended initial implementation. Use one global β_t per temporal block together with a context-dependent $\mathbf{q}_t(x)$. Example-specific persistence $\beta_t(x)$ should be considered only after the global-persistence version is understood, because otherwise model identification and debugging may become unnecessarily difficult.

4 Research Hypotheses

H1: Adaptive memory scale.

Context-dependent multi-timescale gating outperforms a single global or fixed horizon when the optimal horizon varies across examples or subpopulations.

H2: Adaptive persistence.

A learned β_t matches the low-persistence M5b configuration under abrupt or local drift and the high-persistence M5b configuration under recurring or persistent drift without access to a regime label.

H3: Low stationary downside.

Under stationary or weakly drifting conditions, AMG-TP remains close to expanding-history empirical risk minimization (ERM) and does not introduce material calibration loss.

H4: Real CTR benefit.

AMG-TP improves chronological log loss and calibration on at least one real CTR benchmark relative to the best fixed-window and adaptive-window baselines.

H5: Downstream value.

When all predictors feed the same standard bidder, improved CTR prediction yields higher realized value at matched spend.

5 Experimental Plan

5.1 Baselines

The baseline suite should include:

- expanding-history ERM;
- rolling windows of 1, 3, 7, and 14 days or blocks;
- a validation-selected fixed horizon;
- fixed exponential forgetting;
- Han Adaptive Rolling Window (primary adaptive baseline);
- Differentiable Forgetting;
- AdaMoE;
- SFTL as a secondary or negative baseline, without additional tuning budget unless required for reproduction;

- M1 global short/long mixing;
- M2 context-dependent short/long gating;
- M5b default low-smoothness gating;
- M5b high-smoothness gating; and
- Ensemble3 as a diagnostic upper-level competitor rather than necessarily the preferred final method.

5.2 Synthetic Shift Suite

Table 2: Synthetic temporal-shift scenarios.

Scenario	Construction	Primary question
S0 Stationary	No shift	Measure unnecessary adaptation and stationary downside.
S1 Abrupt global	Single global change point	Test rapid response; low β_t should emerge.
S2 Gradual	Parameters drift continuously	Test smooth adaptation.
S3 Recurring	$A \rightarrow B \rightarrow A$	Test retention and persistence; high β_t should emerge.
S4 Local	Only subgroup A changes	Test context-dependent horizon allocation.
S5 Opposing local	Groups change at different times and/or in different directions	Stress-test the assumptions behind a global persistence parameter.
S6 Mixed unknown	Random sequence of shift types	Test whether β_t generalizes without regime labels.

Seeds and confirmation. Use at least 10 seeds for lightweight synthetic experiments. Hyperparameters should be tuned on a development seed set, and confirmatory results should be reported on disjoint seeds.

5.3 Real CTR Evaluation

- Use strict chronological splits; do not mix train and test observations randomly across time.
- Use the current Criteo pipeline as the primary real benchmark. If natural drift remains weak, treat it primarily as a robustness and no-downside test.
- Add a second chronological CTR or recommendation dataset with stronger natural temporal shift if the method’s gains occur mainly on synthetic regimes.
- Because the current locked test has already informed method development, create a fresh confirmation split or use a second real dataset before making final paper-level claims.

5.4 Primary Prediction Metrics

- mean chronological log loss (primary);
- Brier score;
- PR-AUC;
- ROC-AUC (secondary);
- calibration or reliability error; and
- per-day or per-block loss curves.

5.5 Adaptation Metrics

- recovery time after known synthetic shifts;
- peak post-shift excess loss relative to an oracle comparator;
- cumulative post-shift regret;
- stationary downside relative to expanding-history ERM;
- shifted-group and stable-group losses under local drift;
- gate movement $\|\pi_t - \pi_{t-1}\|$ and inferred β_t over time; and
- effective temporal horizon, for example

$$h_t^{\text{eff}}(x) = \sum_k \pi_{t,k}(x) h_k \quad (8)$$

for finite-window experts.

5.6 Oracle Diagnostics

The following quantities should be used diagnostically to quantify remaining headroom rather than as deployable comparators:

- best fixed horizon in hindsight;
- per-period oracle horizon h_t^* ;
- per-group oracle horizon $h_{t,g}^*$;
- oracle fixed smoothness selected from the low/high-smoothness grid;
- oracle per-period persistence β_t^* or smoothness regime; and
- the gap between AMG-TP and each oracle.

Table 3: Ablation suite for identifying the sources of AMG-TP performance.

ID	Variant	Purpose
A0	Expanding only	No temporal adaptation.
A1	M5b, $\beta = 0$ / no persistence	Test the raw gate alone.
A2	Fixed low persistence	Reference for abrupt or local drift.
A3	Fixed high persistence	Reference for recurring drift; analogue of current M5b-high-smooth.
A4	Adaptive β_t , global	Main proposed persistence mechanism.
A5	Adaptive β_t without context-dependent $\mathbf{q}$	Test whether persistence alone explains the gains.
A6	Adaptive β_t plus context-dependent $\mathbf{q}$	Full AMG-TP.
A7	Remove past loss/shift features	Test which state inputs matter.
A8	2 experts versus 5 experts	Test whether a multiscale expert bank is necessary.
A9	Static 5-way mixture / uniform average	Control for model-count advantage.

6 Required Ablations

7 Statistical and Evaluation Discipline

- Use identical temporal episodes or seeds for paired comparisons.
- Freeze hyperparameters on development data before confirmatory runs.
- Report mean effects and uncertainty rather than only the best seed.
- For real chronological data, use paired block-level or day-level bootstrap intervals where appropriate.
- Inspect per-day improvements to ensure aggregate gains are not driven by a small number of unusual days.
- Report parameter count and computational cost so improvements cannot be attributed solely to a larger ensemble.
- Keep all expert predictions causal: each horizon model may use only information available before the prediction block.
- Unit-test causality by verifying that perturbing future labels does not change earlier predictions.

8 Downstream Autobidding Evaluation

The prediction model should be frozen before its economic impact is evaluated. A new bidding policy should not be introduced in the same experiment.

- Feed every CTR model into the same established bidding and pacing policy.

- Use identical budgets and auction episodes for paired comparisons.
- Use realized value at matched spend as the primary downstream metric.
- Also report spend, budget violation, conversion value, and the value-spend frontier.
- Attribute differences to the prediction and adaptation method rather than to bidder changes.

The intended causal chain is

$$\begin{aligned} \text{adaptive temporal training} &\rightarrow \text{better CTR prediction} \\ &\rightarrow \text{same bidder} \rightarrow \text{better value at matched spend.} \end{aligned} \tag{9}$$

9 Success Criteria and Go/No-Go Decisions

Table 4: Decision criteria for interpreting the experimental outcome.

Outcome	Decision rule
Strong success	AMG-TP matches or exceeds low-smooth M5b under abrupt/local drift, matches or exceeds high-smooth M5b under recurring drift, and improves real chronological CTR without material stationary downside.
Partial success	Adaptive β_t improves mixed/unknown regimes but not every specialized regime. Continue if one model replaces multiple hand-tuned specialists with little loss.
Negative result	Fixed high or low persistence dominates adaptive β_t after fair tuning. Retain the empirical smoothness tradeoff as a finding, but do not claim a new adaptive-persistence algorithm.
Insufficient real evidence	Synthetic results are strong but Criteo remains flat. Add another real temporal dataset before making broad machine-learning claims.

10 Recommended Implementation Order

1. Freeze and reproduce all current baseline, M2, M5b, and high-smooth results from one configuration pipeline.
2. Implement AMG-TP with global β_t and the existing M5b raw gate $\mathbf{q}_t(x)$.
3. Run stationary, abrupt, recurring, and local synthetic regimes first.
4. Add mixed/unknown regime training to test whether β_t learns the specialization tradeoff without labels.
5. Run the required ablations and oracle persistence/smoothness diagnostics.
6. Freeze the method and run fresh-seed confirmation.
7. Run a fresh real-data confirmation split or a second dataset.
8. Only then run autobidding impact experiments.

11 Project Summary

AMG-TP is a sequential adaptive-training method for CTR prediction under temporal distribution shift. It maintains predictors trained on multiple historical horizons, uses a context-dependent gate to determine which timescales each incoming example should trust, and learns a separate persistence parameter controlling how quickly those temporal preferences may change. The method is designed to combine the rapid response of short-memory models with the stability and recurrence benefits of long-memory models. Its central empirical test is whether one causally deployed adaptive model can recover the advantages of low-persistence gating under abrupt or local drift and high-persistence gating under recurring drift, and subsequently translate improved CTR prediction into better autobidding value under an unchanged bidding policy.